BANGALORE INSTITUTE OF TECHNOLOGY
K R ROAD, V V PURAM, BANGALORE-04
DEPARTMENT OF ROBOTICS AND ARTIFICIAL INTELLIGENCE
QUESTION BANK 2026-27 (ODD)
Batch: 2024-2028

Course Title: Research Methodology and IPR
Semester: 5th
No. of Lecture Hrs./Week: 03
Total No. of Lecture Hrs.: 40

Course Code: BRMK507
Exam Hours: 03
Exam Marks: 50
Internal Marks:50

Module 1

1. Define the term research and explain the research flow cycle with a relevant diagram.
2. What are the three broad categories of developing and accessing knowledge in research? Explain with a diagram.
3. What are three ways to credit the research contributions? Explain
4. What are the objectives of Engineering Research.
5. Explain the importance of motivation in engineering research.
6. Discuss the different types of engineering research. Clearly point out the differences between all of them with examples.
7. List the different types of Research misconduct? Provide examples for each.
8. Give a detailed description of Ethics and Ethical practices in Research.
9. What are the key ethical issues related to authorship? Explain each one.
10. Differentiate between basic research and applied research in engineering.
11. Explain the different types of research misconduct and their consequences.

Module 2

1. What are the primary goals of conducting a literature review in academic research?
2. How does the new and existing knowledge can contribute to the research process? Explain with relevant points.
3. What are the key features of the bibliographic database of the Web of Science (WoS), and how is it commonly used in research?
4. What are datasheets and write their contents?
5. Enumerate the steps in analyzing literature with the help of a simple literature survey grid.
6. Explain the term citation? Describe the functions of citation?
7. Explain how knowledge flows through a citation network using a flow diagram.
8. Illustrate using a flowchart, how collaboration in a Co-authorship network can improve the flow of knowledge in the research.
9. Explain the most common styles for citation used by engineers during research and provide an example.
10. Explain critical and creative reading techniques and their importance in technical research.
11. Discuss the role of bibliographic databases, Web of Science, Google, and Google Scholar in literature searching.
12. Discuss effective methods for taking notes while reading research papers, including reading mathematics and algorithms.

Module 3

1. What is Intellectual Property Rights (IPR)? Explain the different types of IPR with suitable examples.
2. Discuss the role of Intellectual Property (IP) in the economic and cultural development of society.
3. Discuss the evolution of Intellectual Property and the history of IP protection in India.
4. Explain the process of patenting with flowchart and state the conditions required for obtaining patent protection.
5. Discuss the major amendments in Intellectual Property laws and Acts in India and their significance.
6. Discuss which inventions can be patented and identify the types of subject matter that are not eligible for patent protection.
7. Discuss the feasibility of obtaining worldwide patent protection and justify your explanation with reference to international patent systems.
8. Discuss the feasibility of obtaining worldwide patent protection and justify your explanation with reference to international patent systems.
9. Describe the step-by-step process involved in filing a patent application and represent it through a flowchart.
10. Discuss the importance of utility models in promoting innovation among small-scale inventors.
11. Describe the various kinds of patent applications and their purposes.